Smit Chaudhary

Software Engineer · Quantum Algorithms Developer · Quantum Physicist

Rotterdamm The Netherlands

(+31) 06 1378 2478 smitchaudhary10@gmail.com https://smitchaudhary.github.io/ smitchaudhary 0000-0003-0243-6513

Experience

Quantum Algorithms Developer — Amsterdam, The Netherlands

Pasqal — 2022 - Present

- Developed Quantum Scientific Machine Learning (QSciML) algorithms. Built and trained variational quantum circuits to solve industrially relevant aerodynamics PDEs, creating quantum alternatives to fully classical physics-informed neural networks (PINNs)
- Served as technical lead on European Union funded projects, driving collaboration with major aerospace industry partners
- Translated quantum research into practical solutions for industrial partners, driving business development opportunities
- Architected quantum software infrastructure and built hardware-aware circuit capabilities, making strategic technical decisions that aligned algorithms with neutral atom hardware constraints while enhancing both open-source and proprietary frameworks
- Mentored research interns and junior developers, leading to successful project contributions and published research outcomes

Quantum Computing Intern — California, USA

Menten AI — May 2022 - Sept 2022

- Developed a novel fully quantum generative adversarial network (QGAN) architecture for discrete and binary data generation
- Investigated and implemented quantum circuit features such as noise reuploading in the generator and auxiliary qubits in the discriminator to enhance expressivity
- Demonstrated the model's performance on synthetic datasets and low-energy states of Ising models, showing successful data reproduction and generalization capabilities

Education

Technische Universiteit Delft — Delft, The Netherlands

Master of Science in Applied Physics — August 2020 - July 2022

- **Thesis:** Quantum Walk Based Algorithms for Qubit Mapping (**Supervisor**: Prof. Sebastian Feld)
- **Research Experience:** Honors Project on Noise Induced Barren Plateaus at Leiden University (**Supervisor**: Prof. Jordi Tura)
- **Relevant Coursework:** Applied Quantum Algorithms, Quantum Information, Quantum Computing Architecture

Indian Institute of Technology, Kanpur — Kanpur, India

Bachelor of Science in Physics — July 2016 - May 2020

- **Undergraduate Project:** Bohmian Mechanics and Quantum Information (**Supervisor**: Prof. Kaushik Bhattacharya)
- **Research Experience:** Summer Research at IISER Kolkata on Quantum Machine Learning (**Supervisor**: Prof. Prasanta Panigrahi)
- **Relevant Coursework:** Quantum Computing, Quantum Field Theory, Statistical Mechanics, Probability and Statistics, Optics

Selected Publications

Solving Fluid Dynamics Equations with Differentiable Quantum Circuits
S. Chaudhary, G. T. Balducci, O. Kyriienko, P. K. Barkoutsos, L. Cardarelli, A. A. Gentile
Proceedings of the 35th Parallel CFD International Conference 2024, May 2025

Quantum Circuit Training with Growth-Based Architectures
C. Duffy, **S. Chaudhary**, G. V. Velikova
arXiv preprint, Nov 2024

Towards a scalable discrete quantum generative adversarial neural network
S. Chaudhary, P. Huembeli, I. MacCormack, T. L. Patti, J. Kossaifi, A. Galda
Quantum Science and Technology, April 2023

Skills

Programming	**Python**, **C/C++**, Rust, Verilog
Languages	**English**, Hindi, Gujarati
Libraries & Utilities	**Git**, **PyTorch**, Pennylane, Tensorflow, Jax, Cirq, Qiskit